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Subject : Python

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Project 1:Mini Library Management (Borrow/Return)

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1. Project Title

Borrowing and Returning Book using Python

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2. Problem Statement

To design and implement a Library Management System using Python that allows users to manage library books by performing operations such as viewing available books, borrowing books, returning borrowed books, and displaying the list of borrowed books. The system should use functions to organize these operations and ensure that a book can only be borrowed if it is available in the library and returned only if it was previously borrowed.

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3. Objective

- To create a program that manages a Library Management System for borrowing and returning books.
- To use functions to perform different operations such as displaying books, borrowing books, and returning books.
- To use lists to store available books and borrowed books.
- To apply conditional statements to check whether a book is available for borrowing or eligible for returning.
- To allow users to view available books, borrow books, and return books through a simple menu-driven program.

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5. Description

This project implements a Library Management System using Python that allows users to manage the borrowing and returning of books. The program maintains a list of available books in the library and another list for borrowed books.

Users can view the available books, borrow a book if it is present in the library, and return a book if it was previously borrowed. The system uses functions to perform these operations

and conditional statements to verify whether a book is available for borrowing or valid for returning.

The program is menu-driven, allowing users to select different options repeatedly until they choose to exit. This helps in organizing library book records and demonstrates the use of functions, lists, loops, and conditional statements in Python.

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6. **Algorithm**

1. Start the program.
2. Initialize a list of available books in the library.
3. Create an empty list to store borrowed books.
4. Display a menu with options:
    a) View available books
    b) Borrow a book
    c) Return a book
    d) View borrowed books
    e) Exit
5. Ask the user to enter their choice.
6. If the user selects View available books, display the list of books in the library.
7. If the user selects Borrow a book, ask the user to enter the book name.
8. Check if the entered book is available in the library list.
9. If the book is available, remove it from the library list and add it to the borrowed books list, then display a success message.
10. If the book is not available, display a message that the book is not available.
11. If the user selects Return a book, ask the user to enter the book name.
12. Check if the book exists in the borrowed books list.
13. If it exists, remove it from the borrowed list and add it back to the library list, then display a return success message.
14. If the book is not in the borrowed list, display a message that the book was not borrowed.
15. Repeat the menu until the user chooses the Exit option.
16. If the user selects Exit, terminate the program.

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7. Program

```
library_books = ["Python", "Java", "C++", "HTML", "Data Science"]

borrowed_books = []

def show_books():

    print("\nAvailable Books:")

    for book in library_books:

        print(book)

def show_borrowed():
```

```
    print("\nBorrowed Books:")
    if borrowed_books:
        for book in borrowed_books:
            print(book)
    else:
        print("No books borrowed.")
def borrow_book(book_name):
    if book_name in library_books:
        library_books.remove(book_name)
        borrowed_books.append(book_name)
        print(book_name, "borrowed successfully.")
    else:
        print("Book not available.")
def return_book(book_name):
    if book_name in borrowed_books:
        borrowed_books.remove(book_name)
        library_books.append(book_name)
        print(book_name, "returned successfully.")
    else:
        print("This book was not borrowed.")
while True:
    print("\n----- Library Menu -----")
    print("1. Show Available Books")
    print("2. Borrow Book")
    print("3. Return Book")
    print("4. Show Borrowed Books")
    print("5. Exit")
    choice = input("Enter your choice: ")
    if choice == "1":
        show_books()
```

```
    elif choice == "2":
        book = input("Enter book name to borrow: ")
        borrow_book(book)
    elif choice == "3":
        book = input("Enter book name to return: ")
        return_book(book)
    elif choice == "4":
        show_borrowed()
    elif choice == "5":
        print("Exiting program...")
        break
    else:
        print("Invalid choice.")
```

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8. Sample Output

```
Output                                                    Clear
----- Library Menu -----
1. Show Available Books
2. Borrow Book
3. Return Book
4. Show Borrowed Books
5. Exit
Enter your choice: 1

Available Books:
Python
Java
C++
HTML
Data Science

----- Library Menu -----
1. Show Available Books
2. Borrow Book
3. Return Book
4. Show Borrowed Books
5. Exit
Enter your choice: 2
Enter book name to borrow: Python
Python borrowed successfully.

----- Library Menu -----
1. Show Available Books
2. Borrow Book
3. Return Book
4. Show Borrowed Books
5. Exit
Enter your choice: 2
Enter book name to borrow: HTML
HTML borrowed successfully.

----- Library Menu -----
1. Show Available Books
2. Borrow Book
3. Return Book
4. Show Borrowed Books
5. Exit
Enter your choice: 3
Enter book name to return: HTML
HTML returned successfully.
```

```
----- Library Menu -----
1. Show Available Books
2. Borrow Book
3. Return Book
4. Show Borrowed Books
5. Exit
Enter your choice: 4

Borrowed Books:
Python

----- Library Menu -----
1. Show Available Books
2. Borrow Book
3. Return Book
4. Show Borrowed Books
5. Exit
Enter your choice: 1

Available Books:
Java
C++
Data Science
HTML

----- Library Menu -----
1. Show Available Books
2. Borrow Book
3. Return Book
4. Show Borrowed Books
5. Exit
Enter your choice: 5
Exiting program...

=== Code Execution Successful ===
```

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9. Result

The Library Management System successfully allows users to view available books, borrow books, and return borrowed books using functions. The program correctly updates the lists of available and borrowed books and displays appropriate messages based on the user's actions.

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10. Conclusion

This project demonstrates how Python can be used to implement a simple Library Management System using functions. It shows the use of lists, loops, and conditional statements to manage library operations such as viewing, borrowing, and returning books. The program provides a clear example of how Python can be applied to solve real-world problems related to managing records and organizing data efficiently.